

BAREC-ST-2026: The Second Shared Task on Arabic Readability Assessment

Khalid N. Elmadani^{1,2} Bashar Alhafni¹ Hanada Taha-Thomure³ Nizar Habash²

¹Mohamed bin Zayed University of Artificial Intelligence

²New York University Abu Dhabi

³Zayed University

{khalid.elmadani, bashar.alhafni}@mbzuai.ac.ae

nizar.habash@nyu.edu, hanada.thomure@zu.ac.ae

Abstract

We present the results and findings of the BAREC Shared Task 2026, the second shared task on Arabic readability assessment, organized as part of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026). The shared task focuses on automatic readability assessment using the BAREC Corpus (Elmadani et al., 2025b), addressing fine-grained classification across 19 readability levels. It comprises two subtasks—sentence-level classification and document-level classification—and three tracks: (1) the *Strict Track*, which permits only the BAREC Corpus; (2) the *Constrained Track*, which permits the BAREC Corpus, the SAMER Corpus (Alhafni et al., 2024), and the SAMER Lexicon (Al Khalil et al., 2020); and (3) the *Open Track*, which permits the use of any external resources. A total of 14 teams from 12 countries registered for the task, of which 10 submitted system description papers. All participating teams competed only in the sentence-level classification subtask. The top-performing system achieved a quadratic weighted kappa (QWK) of 85.3.¹

1 Introduction

Readability assessment plays an important role in education, literacy development, and language learning by helping ensure that texts are appropriate for readers’ proficiency levels. Texts that are poorly matched to readers’ abilities may adversely affect comprehension, retention, reading speed, and engagement (DuBay, 2004; Klare, 1963). Consequently, text-leveling systems are widely used, particularly in early education, to support structured and measurable progress in reading development (Allington et al., 2015; Barber and Klauda, 2020).

Although readability models have been developed for several languages, substantial challenges

RL	Grade	Example
1	KG	Sara سارة
3	1st	An egg in the nest! بيضة في العش!
7	2nd	هل جئت بصيد وفير؟ هل جئت بصيد وفير؟
10	4th	Did you come back with a good catch? كان يجري مفزوعاً وكأنه يسابق الريح!
		He was running in fright, as though racing the wind!
14	8th	الأبعاد الجيوسياسية لندرة المياه
		The geopolitical dimensions of water scarcity
17	Uni	حيا وقال درتان في وعاء ممتن
		He greeted [them] and said, “Two pearls in a humble vessel.”

Table 1: Examples by Reading Level (RL) and grade.

remain, especially for fine-grained text leveling and resource-scarce languages. For example, Fountas and Pinnell’s 27-level framework for English (Fountas and Pinnell, 2006) and Taha-Thomure’s (2017) 19-level framework for Arabic illustrate the value of detailed readability classifications. Such fine-grained levels enable more precise educational applications while remaining flexible enough to be mapped to broader categories when needed.

The Taha/Arabi21 framework (Taha-Thomure, 2017), which has been used to annotate more than 9,000 children’s books, forms the basis of our work. Building on this framework, the BAREC guidelines (Habash et al., 2025) provide a standardized approach to sentence-level readability assessment across a broad range of genres and educational stages, from early childhood through postgraduate education (see Table 1). For the complete guidelines in Arabic and English, we refer readers to Habash et al. (2025).

Arabic presents distinctive challenges for readability assessment because of its rich morphology, extensive lexicon, and highly ambiguous orthography. In contrast to English, for which established readability formulas and datasets are available, Ara-

¹<https://barec.camel-lab.com/sharedtask2026>

bic readability research has long lacked standardized resources. This gap has limited the development of robust computational models that can accurately assess text difficulty across proficiency levels.

The BAREC Shared Task 2026 was organized as part of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026), co-located with EMNLP 2026. Fourteen teams from 12 countries registered for the shared task, and 10 teams submitted system description papers, which are cited throughout this overview (see §5). This paper describes the task design, summarizes the submitted systems, and presents the official results.

The paper is structured as follows: §2 reviews related work. §3 outlines the sub-tasks and tracks of the shared task. §4 introduces the datasets and evaluation metrics. §5 describes the baselines and provides an overview of the submitted systems. Finally, §6 reports and discusses the results.

2 Related Work

Automatic Readability Assessment Automatic readability assessment has been studied extensively across languages, yielding a wide range of datasets, features, and modeling approaches (Collins-Thompson and Callan, 2004; Pitler and Nenkova, 2008; Feng et al., 2010; Vajjala and Meurers, 2012; Xu et al., 2015; Nadeem and Ostendorf, 2018; Vajjala and Lučić, 2018; Deutsch et al., 2020; Lee et al., 2021). Early work, particularly for English, often relied on textbook collections, whose graded organization naturally supports readability evaluation (Vajjala, 2022). More recent work has expanded to alternative sources, including crowdsourced readability annotations from online platforms (Vajjala and Meurers, 2012; Vajjala and Lučić, 2018) and proficiency examinations designed for second-language learners under the CEFR framework (Xia et al., 2016). These efforts have also motivated increasingly fine-grained approaches to readability prediction, supporting more precise educational applications.

Arabic Readability Efforts Arabic readability research has evolved from handcrafted formulas based on surface features, such as word and sentence length (Al-Dawsari, 2004; Al-Khalifa and Al-Ajlan, 2010; Hazim et al., 2022), to machine-learning approaches that incorporate linguistic features, including part-of-speech tags, morphological information, diacritization, and script-specific

characteristics (Forsyth, 2014; Saddiki et al., 2018; El-Haj and Rayson, 2016). Recent work has further demonstrated the effectiveness of pretrained language models for Arabic readability assessment (Liberato et al., 2024). Several graded resources have also been developed. The Taha/Arabi21 framework (Taha-Thomure, 2017) defines 19 readability levels for children’s books and has been used to level more than 9,000 books. Other efforts include CEFR-based resources and vocabulary profiling (Habash and Palfreyman, 2022; Naous et al., 2024; Soliman and Familiar, 2024), the DARES datasets derived from Saudi school materials (El-Haj et al., 2024; Almujaivel et al., 2025), and the SAMER project, which introduced a five-level readability lexicon (Al Khalil et al., 2020) and a manually annotated Arabic text simplification corpus (Alhafni et al., 2024).

BAREC Project This shared task is part of the broader BAREC project, which develops resources and tools for fine-grained Arabic readability assessment across the Arab world. The project is grounded in the BAREC Annotation Framework, which provides 19-level guidelines for Arabic readability annotation (Habash et al., 2025). Building on these guidelines, the BAREC Corpus provides a large-scale, manually annotated dataset for sentence-level readability assessment (Elmadani et al., 2025b), while BAREC Demo offers a web-based interface for assessing the readability of Arabic sentences (Altarbouch et al., 2025). The first BAREC Shared Task (Elmadani et al., 2025a) established a community benchmark for Arabic readability assessment, and the BAREC-10M Corpus subsequently extended the available data with a substantially larger automatically labeled corpus (Elmadani et al., 2026). The BAREC Shared Task 2026 builds on this foundation by providing a new evaluation setting for fine-grained sentence- and document-level Arabic readability classification.

History of BAREC Shared Task This is the second edition of the BAREC Shared Task. The first edition, BAREC Shared Task 2025, was organized as part of the Third Arabic Natural Language Processing Conference (ArabicNLP 2025), co-located with EMNLP 2025. It attracted 22 registered teams from 12 countries, 17 of which submitted system description papers. Most participating teams relied on pretrained language models (PLMs), often augmented with morphological features; several systems additionally used ensembling, label bal-

Split	#Documents	#Sentences	#Words
Train	1,518 (79%)	54,845 (79%)	832,743 (80%)
Dev	194 (10%)	7,310 (11%)	101,364 (10%)
Test	210 (11%)	7,286 (10%)	105,264 (10%)
All	1,922 (100%)	69,441 (100%)	1,039,371 (100%)
ST-2025	100	3,420	53,052
ST-2026	212	8,077	110,981

Table 2: **BAREC** Corpus splits.

ancing, or ordinal-aware loss functions. Teams such as !MSA (Basem et al., 2025), AMAR (Saeed et al., 2025), and Qais (Ahmed, 2025) also explored large language models (LLMs), whereas GNNinjas (Elchafei et al., 2025) and Syntaxa (Bahloul, 2025) investigated graph neural networks (GNNs) (Zhou et al., 2021), and Pixel (Sapirstein, 2025) explored vision-language models (Dosovitskiy et al., 2021; Rust et al., 2023). Traditional non-neural approaches were comparatively rare. Overall, the most successful systems combined PLMs with linguistic features and ensemble-based methods, establishing a strong baseline for subsequent BAREC shared-task editions.

The test set used in this edition is approximately twice the size of that used in the previous shared task (see Table 2). To better assess system generalization, it includes data from 17 publishing houses that were not represented in the original BAREC Corpus, enabling a more robust evaluation of a system’s ability to handle previously unseen text sources.

3 Task Description

The BAREC Readability Assessment Shared Task evaluates systems for fine-grained Arabic readability classification under a 19-level framework. Participants were invited to develop models that predict readability at both the sentence and document levels.

Sub-tasks The shared task comprises the following two subtasks:

1. **Sentence-Level Classification:** Given an individual sentence, predict its readability level.
2. **Document-Level Classification:** Given a document consisting of consecutive sentences, predict its readability level. The document label is defined by the readability level of its most difficult sentence.

Tracks Each sub-task is offered under three tracks with different constraints on the resources that participants may use:

- **Strict Track:** Systems may be trained exclusively on the BAREC Corpus (Elmadani et al., 2025b).
- **Constrained Track:** Systems may use the BAREC Corpus, the SAMER Corpus (Alhafni et al., 2024) and the SAMER Lexicon (Al Khalil et al., 2020).
- **Open Track:** Systems may use any publicly available external resources.

Together, the two sub-tasks and three tracks define six possible subtask–track combinations. Participants were permitted to submit to multiple combinations.

4 Shared Task Datasets and Evaluation

This section describes the datasets available under the different tracks, the evaluation metrics, and the submission guidelines provided to participants.

4.1 Datasets

BAREC Corpus The BAREC Corpus (Elmadani et al., 2025b) is the primary dataset used in the shared task. It contains 1,922 documents and 69,441 sentences annotated with 19 readability levels. The corpus is partitioned at the document level into **Train** ($\simeq 80\%$), **Dev** ($\simeq 10\%$), and **Test** ($\simeq 10\%$) splits. Table 2 presents statistics for these splits at the document, sentence, and word levels.

SAMER Corpus The SAMER Corpus (Alhafni et al., 2024) contains 4,289 documents (158K words) and 20,358 fragments annotated with three readability levels. For this shared task, we use the fragment data made available and reported by Liberatoro et al. (2024). Table 4 summarizes the corpus statistics for the Train, Dev, and Test splits.

SAMER Lexicon The SAMER Lexicon (Al Khalil et al., 2020) is a readability lexicon for Modern Standard Arabic (MSA) containing 40K lemma–part-of-speech pairs annotated with five readability levels. The lexicon was manually annotated by three language professionals from different regions of the Arab world. Table 5 presents the distribution of readability levels in the lexicon.

				BAREC Corpus v1 (Sentences)											
Level-3	Level-5	Level-7	Level-19	All		Train		Dev		Test		ST-2025		ST-2026	
1	1	1	1-alif	409	1%	333	1%	44	1%	32	0%	21	1%	37	0%
			2-ba	437	1%	333	1%	68	1%	36	0%	21	1%	98	1%
			3-jim	1,462	2%	1,139	2%	182	2%	141	2%	69	2%	194	2%
			4-dal	751	1%	587	1%	78	1%	86	1%	28	1%	61	1%
		2	5-ha	3,443	5%	2,646	5%	417	6%	380	5%	188	5%	325	4%
			6-waw	1,534	2%	1,206	2%	189	3%	139	2%	47	1%	185	2%
			7-zay	5,438	8%	4,152	8%	701	10%	585	8%	296	9%	647	8%
	2	3	8-Ha	5,683	8%	4,529	8%	613	8%	541	7%	263	8%	775	10%
			9-ta	2,023	3%	1,597	3%	236	3%	190	3%	101	3%	310	4%
		4	10-ya	9,763	14%	7,741	14%	1,012	14%	1,010	14%	457	13%	1,547	19%
			11-kaf	4,914	7%	4,041	7%	409	6%	464	6%	233	7%	479	6%
2	3	5	12-lam	14,471	21%	11,318	21%	1,491	20%	1,662	23%	682	20%	1,713	21%
			13-mim	4,039	6%	3,252	6%	349	5%	438	6%	177	5%	405	5%
3	4	6	14-nun	10,687	15%	8,573	16%	1,072	15%	1,042	14%	596	17%	1,048	13%
			15-sin	2,547	4%	2,016	4%	258	4%	273	4%	171	5%	174	2%
	5	7	16-ayn	1,141	2%	866	2%	114	2%	161	2%	55	2%	49	1%
			17-fa	480	1%	364	1%	49	1%	67	1%	15	0%	23	0%
			18-sad	103	0%	67	0%	13	0%	23	0%	0	0%	0	0%
			19-qaf	116	0%	85	0%	15	0%	16	0%	0	0%	7	0%
Total				69,441	100%	54,845	100%	7,310	100%	7,286	100%	3,420	100%	8,077	100%

Table 3: Sentence-level splits across BAREC readability levels.

Split	#Documents	#Fragments	#Words
Train	2,790 (65%)	14,256 (70%)	112,828 (71%)
Dev	607 (14%)	2,948 (14%)	22,075 (14%)
Test	892 (21%)	3,154 (15%)	23,161 (15%)
All	4,289 (100%)	20,358 (100%)	158,064 (100%)

Table 4: SAMER Corpus splits.

Level	Type Count
Level I	3,545 (9%)
Level II	3,221 (8%)
Level III	5,510 (14%)
Level IV	10,130 (25%)
Level V	18,281 (45%)
Total	40,687 (100%)

Table 5: SAMER Lexicon distributions

BAREC-ST-2026 Blind Test Set As in the BAREC Shared Task 2025, we provide a new blind test set for the final evaluation, annotated according to the 19-level BAREC framework. This test set is approximately twice the size of that used in the previous edition. To better assess system generalization, it includes texts from 17 publishing houses that were not represented in the original BAREC Corpus. This design enables a more robust evaluation of systems’ ability to handle text from

previously unseen sources.

4.2 Evaluation Metrics

Following Elmadani et al. (2025b), we formulate readability assessment as an ordinal classification problem and evaluate submitted systems using the following metrics:

- **Accuracy (Acc):** The proportion of predictions that exactly match the reference labels under the 19-level scheme (Acc^{19}). We additionally report coarse-grained accuracy measures (Acc^7 , Acc^5 , and Acc^3) obtained by collapsing the 19 levels into 7, 5, and 3 levels, respectively (see Table 3).
- **Adjacent Accuracy ($\pm 1 \text{ Acc}^{19}$):** Also known as off-by-one accuracy, this metric considers a prediction correct when it exactly matches the reference label or differs from it by one readability level.
- **Average Distance (Dist):** Equivalent to mean absolute error (MAE), this metric measures the average absolute difference between predicted and reference labels.
- **Quadratic Weighted Kappa (QWK):** An extension of Cohen’s kappa (Cohen, 1968; Doewes et al., 2023), QWK measures agree-

ment between predicted and reference labels while assigning quadratically greater penalties to errors that are farther from the reference label.

As in the first BAREC Shared Task, we use **QWK** as the primary metric for ranking systems. We prioritize QWK because it captures the ordinal structure of readability levels and provides a smooth, distance-sensitive penalty for misclassifications, unlike the hard decision boundaries used by accuracy-based measures.

4.3 Submission Guidelines

The shared task was organized in two phases: a development phase and a testing phase. During the **development phase**, we hosted CodaBench (Xu et al., 2022) competitions for all tracks. Participants could evaluate their systems locally using the BAREC development and test sets,² or submit predictions for the BAREC Test set through the corresponding CodaBench competition. Because the BAREC Test set is publicly available, this phase was open to all interested participants.

During the **testing phase**, the official blind test set was released exclusively to registered participants. Registered teams were required to submit a system description paper.

5 Participants and Systems

Table 6 summarizes the main modeling approaches adopted by the participating teams. All submitted systems fine-tuned pretrained language models, often combining them with ordinal-aware loss functions, custom decision thresholds, ensemble methods, and linguistic or surface-level features. A general trend across submissions is that systems exploring and integrating a broader range of complementary techniques tended to achieve stronger performance. In this section, we provide a brief description of each team’s approach, followed by several baselines.

Thylinao (Silchenko, 2026) Thylinao participated in the sentence-level Strict and Open tracks, submitting the same system to both. They used an equal-weight ensemble of 14 fine-tuned encoder models spanning several pretrained Arabic and multilingual language models, loss functions, and input representations, including raw text, word-level input, and D3Tok morphological segmentation. The

member scores were standardized before averaging, and the final predictions were produced using custom cutpoints placed based on validation-optimized thresholds. They achieved a QWK score of 85.3 and ranked first in both tracks.

ZHR (Zaher, 2026) ZHR participated in the sentence-level Strict and Open tracks. Their systems combined several pretrained Arabic encoders fine-tuned using five ordinal objectives, with CAMEL-Tools morphological segmentation (Obeid et al., 2020), greedy ensemble selection, and QWK-optimized decision thresholds. For the Strict Track, the team used target-prior calibration to account for label-distribution shift in the development set, achieving a QWK score of 85.0. For the Open Track, they additionally pretrained selected models on silver-labeled data from BAREC-10M (Elmadani et al., 2026), achieving a QWK score of 85.2.

Nile Nexus (Zaman and Rishta, 2026) Nile Nexus participated in the sentence-level Strict Track. They fine-tuned several Arabic pretrained language models using different ordinal-aware objectives, including soft ordinal cross-entropy, squared Earth Mover’s Distance (L. Hou, 2017), direct ordinal regression, and CORAL-style cumulative-threshold loss (Cao et al., 2020; Shi et al., 2023). They combined the strongest models using validation-based greedy ensemble selection optimized for QWK and applied learned, level-specific decision thresholds to convert continuous scores into readability labels. Their system achieved a QWK score of 84.8, ranking third in the Strict Track.

Skywalker (Liang and Li, 2026) Skywalker participated in the sentence-level Strict Track. They fine-tuned a diverse set of Arabic pretrained language models using classification, regression, Earth Mover’s Distance (L. Hou, 2017), and multitask objectives. They combined model scores, confidence estimates, rank-based features, and metadata through cross-fitted LightGBM components (Ke et al., 2017). The resulting continuous scores were fused and mapped to the 19 readability levels using validation-optimized monotonic thresholds. Their system achieved a QWK score of 84.2, ranking fourth in the Strict Track.

Viper (Khang et al., 2026) Viper participated in the sentence-level Strict Track. They trained

²<https://github.com/CAMEL-Lab/barec-shared-task-2025>

Team	Affiliation	Sentence-Level Tracks	QWK	Ling.	PLM -FT	Ord. Loss	Cust. Thr.	Ens.
Thylinao	NUS, Singapore	Strict, Open	85.3	✓	✓	✓	✓	✓
ZHR	Nile University, Egypt	Strict, Open	85.2	✓	✓	✓	✓	✓
Nile Nexus	CUET, Bangladesh	Strict	84.8		✓	✓	✓	✓
Skywalker	MBZUAI, UAE	Strict	84.2		✓	✓	✓	✓
Viper	UIT, VNU-HCM, Vietnam	Strict	84.1	✓	✓	✓		✓
HP	UIT, VNU-HCM, Vietnam	Strict	83.7		✓	✓	✓	✓
NAMAA	AIU, Syria	Strict	82.4	✓	✓	✓		✓
LSI	LSI, USA	Constrained, Open	81.7		✓			✓
Soloeval	Habib University, Pakistan	Open	80.6	✓	✓	✓		
Northwestern_2	NU-Q, Qatar	Strict	79.0		✓			

Table 6: Summary of the participating teams, their affiliations, sentence-level tracks, best official QWK scores, and reported modeling approaches. QWK scores are reported as percentages; for teams participating in multiple tracks, the highest score is shown. *Ling.* indicates the use of linguistic or surface-level features, *PLM-FT* the fine-tuning of pretrained language models, *Ord. Loss* an ordinal-aware training objective, *Cust. Thr.* custom decision-threshold optimization, and *Ens.* model ensembling. A checkmark denotes that the corresponding technique was used.

Team	Track	QWK	Acc ¹⁹	±1 Acc ¹⁹	Dist	Acc ⁷	Acc ⁵	Acc ³
Thylinao [†]	Strict	85.3	37.7	70.8	1.14	60.5	66.9	74.6
ZHR	Open	85.2	39.6	72.1	1.08	61.4	68.5	75.8
ZHR	Strict	85.0	38.8	72.0	1.10	60.4	68.7	75.8
Nile Nexus	Strict	84.8	37.9	73.5	1.10	61.4	68.4	75.1
Skywalker	Strict	84.2	41.7	73.3	1.10	60.9	69.2	76.1
Viper	Strict	84.1	40.9	73.6	1.10	62.3	69.2	75.7
HP	Strict	83.7	47.7	72.6	1.05	63.5	69.6	76.4
Baseline III	Strict	82.5	42.4	73.6	1.09	61.6	68.8	75.6
NAMAA	Strict	82.4	37.1	73.2	1.10	59.5	67.2	74.7
LSI [‡]	Constrained	81.7	50.6	72.9	1.00	65.0	71.2	77.2
Baseline II	Strict	81.1	59.2	71.8	1.01	68.3	72.5	77.3
soloeval [§]	Strict	80.6	49.1	73.6	1.10	64.5	70.0	75.1
Baseline I	Strict	79.7	58.2	71.1	1.04	66.6	72.4	77.0
Northwestern_2	Strict	79.0	55.6	70.0	1.10	65.3	70.5	76.2

Table 7: Official sentence-level results for the participating teams and organizer baselines, ordered by QWK. [†]The Open Track result is omitted because the same Thylinao system was submitted to both the Strict and Open tracks. [‡]The Open Track result is omitted because the same LSI system was submitted to both the Constrained and Open tracks. [§]Although Soloeval submitted to the Open Track, their system is listed as Strict-equivalent because it used only resources permitted under the Strict Track. All scores except Dist are reported as percentages.

five AraBERTv2-based (Antoun et al., 2020) regression models using paired representations that combined D3Tok morphological segmentation with normalized surface text, sentence-level statistics, and corpus metadata. To mitigate class imbalance, they applied level-aware weighted sampling during training. The five models were trained with different random seeds and combined through uniform averaging, after which the continuous ensemble score was rounded to the nearest readability level. Their system achieved a QWK score of 84.1, ranking fifth in the Strict Track.

HP (Dinh and Van, 2026) HP participated in the sentence-level Strict Track. They applied parameter-efficient fine-tuning with LoRA (Hu et al., 2022) to AraBERTv02 (Antoun et al., 2020)

and ARBERT (Abdul-Mageed et al., 2021), formulating the task as ordinal regression with a sample-weighted CORAL loss. They trained their models using stratified five-fold cross-validation and multiple random seeds. For post-processing, they combined predictions obtained from QWK-optimized Nelder–Mead thresholds with distribution-aligned predictions through an equal-weight blend. Their best system, based on AraBERTv02, achieved a QWK score of 83.7 on the blind test set.

NAMAA (Sibae et al., 2026) NAMAA participated in the sentence-level Strict Track. They evaluated a progression of approaches ranging from engineered linguistic and morphological features and TF–IDF representations to contextual transformer models, multi-encoder fusion, and

output-level ensembling. Their final submission used EmbeddingGemma-300M (Schechter Vera et al., 2025) with Gaussian soft labels, an ordinal KL-divergence objective, inverse-frequency class weighting, concatenated [CLS] and mean-pooled representations, and expected-value decoding. They selected the best checkpoint according to development-set QWK. Their system achieved a QWK score of 82.4 on the blind test set.

LSI (Al-Madi, 2026) LSI participated in the sentence-level Constrained and Open tracks, submitting the same system to both. They built an equal-weight, logit-level ensemble of three complementary Arabic pretrained language models: AraBERT (Antoun et al., 2020), AraModernBERT (Elshehy et al., 2026), and CAMELBERT-Mix (Inoue et al., 2021). AraBERT and AraModernBERT were trained with cross-entropy, whereas CAMELBERT-Mix used an ordinal Earth Mover’s Distance loss and incorporated six sentence-level lexical-difficulty features derived from the SAMER Lexicon. Their ensemble achieved a QWK score of 81.7, ranking first in the Constrained Track and third in the Open Track.

Soloeval (Hussain Usmani, 2026) Soloeval participated in the sentence-level Open Track. They fine-tuned MARBERTv2 (Abdul-Mageed et al., 2021) and CAMELBERT-Mix (Inoue et al., 2021) separately using a soft-label ordinal classification objective, in which Gaussian target distributions centered on the gold levels encouraged distance-sensitive predictions aligned with QWK. They augmented the encoder representations with five surface features capturing sentence length, word count, average word length, diacritic density, and punctuation density. Their final submission used CAMELBERT-Mix with three-fold cross-validation and averaged the predictions of the fold-specific models. Although the Open Track permitted additional resources, they used only the official BAREC data and achieved a QWK score of 80.6.

Northwestern_2 (Shlkamy and Zaghouani, 2026) Northwestern_2 participated in the sentence-level Strict Track. They formulated readability assessment as standard 19-class classification and fine-tuned CAMELBERT-Mix using cross-entropy loss. Sentences were tokenized with the model’s native tokenizer, truncated to 128 subword tokens, and dynamically padded during training. The team selected the best checkpoint

according to development-set QWK, and their submitted system achieved a QWK score of 79.0 on the blind test set.

Baselines We evaluated three baselines for comparison with the participating systems. All three are Arabic-specific BERT-base models fine-tuned on the BAREC Corpus and selected from the models introduced by Elmadani et al. (2025b). The baselines differ along three dimensions:

- **Pretrained language model:** AraBERTv02 or AraBERTv2 (Antoun et al., 2020).
- **Input representation: Word**, which applies basic sentence preprocessing, including the removal of diacritics and kashida; or **D3Tok**, which uses morphological segmentation to separate words into their base and clitic components.³
- **Training objective:** cross-entropy classification (**CE**) or regression with mean squared error (**Reg**).

Based on these dimensions, we selected the following baselines:

- **AraBERTv02+Word+CE (Baseline I):** A standard classification baseline that combines the widely used AraBERTv02 model with word-level input and a cross-entropy objective.
- **AraBERTv2+D3Tok+CE (Baseline II):** A classification baseline designed to measure the effect of linguistically motivated D3Tok segmentation.
- **AraBERTv2+D3Tok+Reg (Baseline III):** A regression baseline motivated by the ordered structure of the readability labels, for which the distance between the predicted and reference levels is meaningful.

6 Results

Table 7 presents the official results on the sentence-level blind test set, together with the baselines. No teams submitted systems for the document-level subtask. For teams that submitted an identical system to multiple tracks, duplicate results are omitted as described in the table caption.

By linking the results in tables 6 and 7, we see that the strongest systems generally combined several complementary techniques. In addition to fine-

³The D3Tok representation is generated using CAMEL Tools (Obeid et al., 2020).

tuning pretrained language models, the highest-ranked approaches typically incorporated ordinal-aware objectives, customized decision thresholds, and model ensembling. Several systems also exploited morphological, linguistic, or surface-level information. This pattern suggests that combining diverse modeling and calibration strategies generally led to better performance. Access to additional resources was also not consistently decisive: the improvement obtained by the leading Open Track system over its Strict counterpart was small.

The results further highlight the distinction between exact classification accuracy and ordinal agreement. Some systems achieved comparatively high exact or coarse-grained accuracy while receiving lower QWK scores, whereas the highest-ranked systems under QWK did not necessarily obtain the best exact accuracy. This indicates that correctly modeling the ordering of the labels and avoiding distant errors are more important for QWK than maximizing exact matches alone. The baseline comparison supports this observation: the regression-based baseline achieved the strongest baseline QWK, while a cross-entropy baseline performed better on several accuracy measures. Overall, these findings reinforce the importance of ordinal-aware training, ensemble diversity, and prediction calibration for fine-grained Arabic readability assessment.

7 Conclusion

We organized the BAREC Shared Task 2026, the second shared task on fine-grained Arabic readability assessment. The task offered sentence- and document-level classification under Strict, Constrained, and Open resource settings, using the 19-level BAREC framework. 14 teams from 12 countries registered, and 10 submitted system description papers. The official blind test set included texts from 17 previously unseen publishing houses, provided a challenging evaluation of cross-source generalization. The strongest systems combined pretrained language models with ordinal-aware objectives, custom thresholds, linguistic features, and ensembling. We hope this shared task encourages further research on robust sentence- and document-level Arabic readability assessment.

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Limitations

This work has several limitations. First, The observed relationship between broader use of ordinal objectives, custom thresholds, linguistic features, and ensembling and stronger performance is correlational: the submitted systems differ in many respects, preventing the contribution of individual components from being isolated. Second, we use Quadratic Weighted Kappa (QWK) as the primary evaluation metric because it accounts for the ordinal nature of the readability levels. Nevertheless, determining the most appropriate metric for fine-grained readability assessment remains an open research question.

Ethics Statement

We used AI writing assistance within the scope of “Assistance purely with the language of the paper” described in the ACL Policy on Publication Ethics.

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